Topology Optimization
Lecture 13

Improving the efficiency of upconversion by light concentration using nanoparticle design

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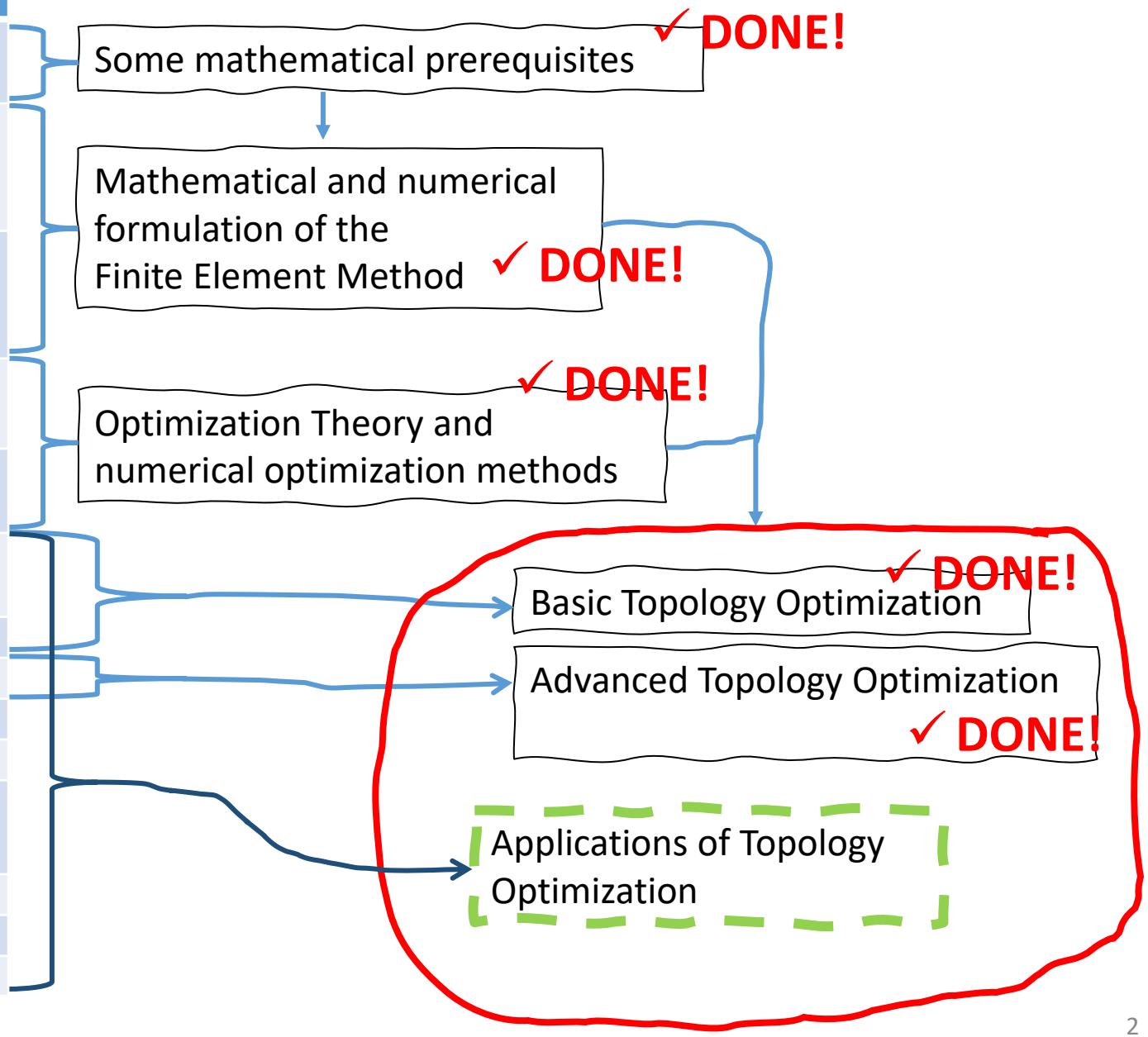
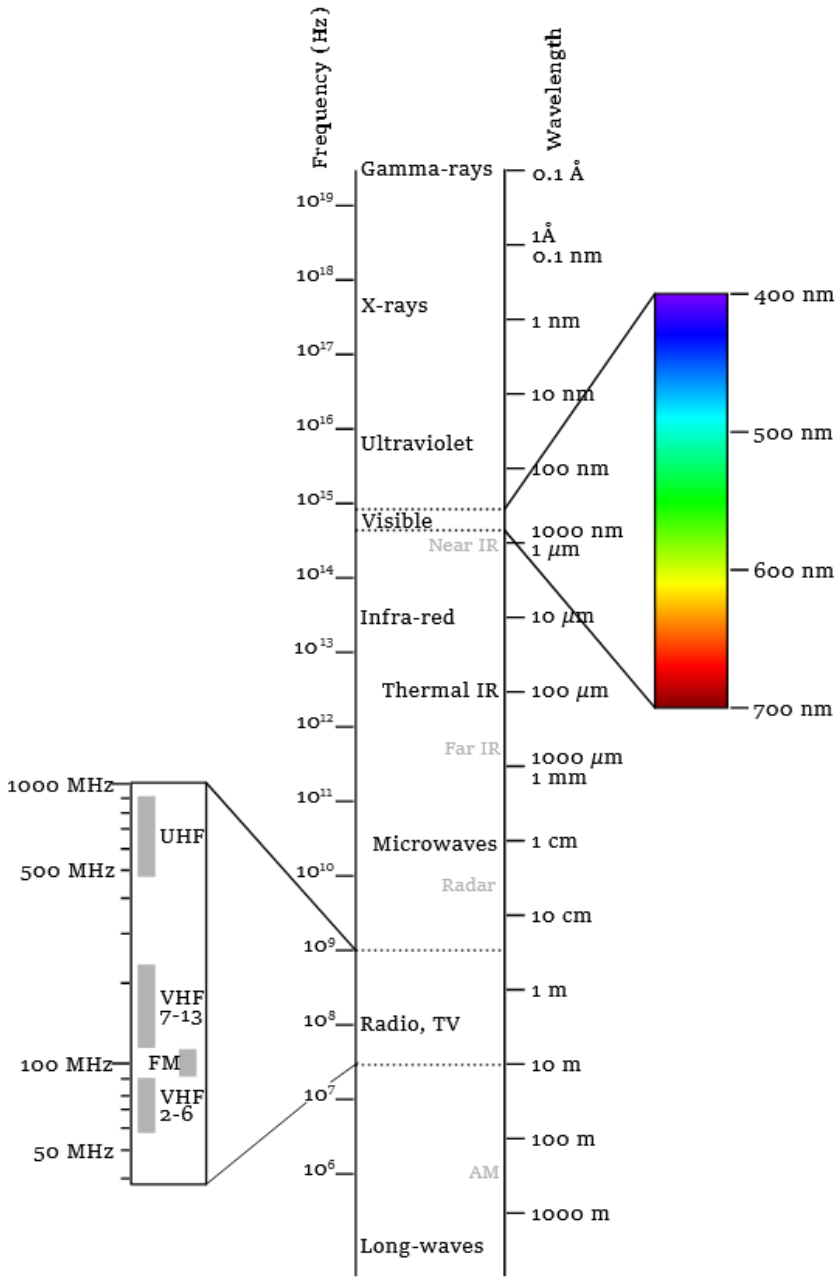
Course overview

No.	Topics
1	Introduction Function and vector spaces, Interpolation, L^2 projection
2	The Finite Element Method for PDEs 1: Poisson's Equation, The Finite Element Method, Properties of the Finite Element Method
3	The Finite Element Method for PDEs 2: Index notation, Linear Elasticity, Non-linear Poisson's equation, Non-linear Elasticity
4	Constrained Optimization 1: Optimality, KKT Theorem, Numerical Optimization Algorithms
5	Constrained Optimization 2: Method of Moving Asymptotes, MPI and PETSc
6	Introduction to Topology Optimization: The idea, Filtering and Projection, Python implementation
7	Gradients
8	Robust topology optimization, More on MPI
9	Wave equations
10	Acoustic waves and more on Robust Optimization
11	Penalty and Augmented Lagrangian Methods Stress-constraints in Topology Optimization 1
12	Stress-constraints in Topology Optimization 2
13	Light concentration using nanoparticle design
14	Heat conduction



Recap: Electromagnetic Waves

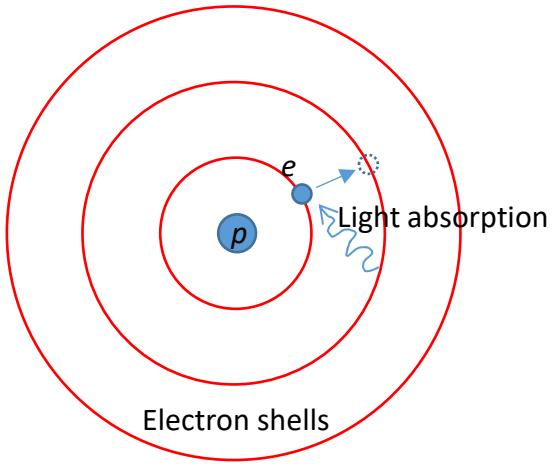
- Light waves are waves in electric and magnetic fields.
- Since electric and magnetic field exist both inside and outside materials, the waves also exist in vacuum (e.g. outer space).
- The origin is in Maxwell's equations in vacuum, which work for length scales from below 0.1 Å to the size of the universe.
- Maxwell's equations in matter generally work well "only" from ~1 nm and above due to the constitutive relations.
- In the linear approximation, the waves are described both inside and outside materials by a linear vector wave equation, whose solutions yield the electric field or magnetic field as a function of space and time.
- The frequencies range from below 1 Hz to 10^{25} Hz in the electromagnetic spectrum.

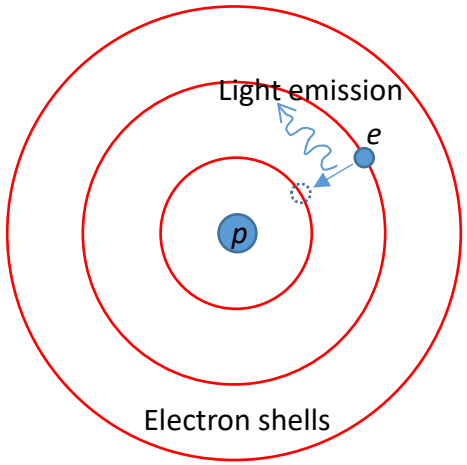


https://en.wikipedia.org/wiki/Electromagnetic_spectrum

Light absorption and emission

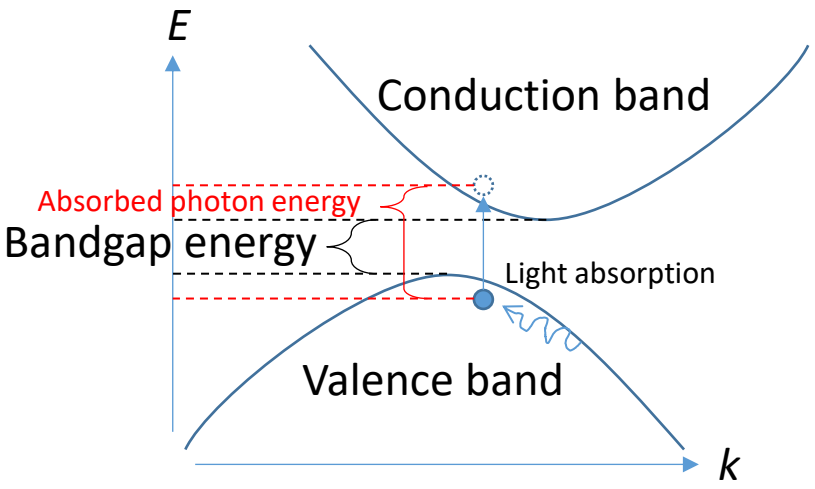
- Atoms can be excited by an external light source.
- They can relax again to their ground state by emission of light/a photon.
- Simplest atom: Hydrogen
  - Only 1 electron.
  - When light is absorbed, the energy of the electron is raised, and the electron is jumping to a higher energy state (usually a different 'shell').
  - When light is emitted, the energy of the electron is lowered, and the electron is jumping to a lower energy state (usually a different 'shell').
- The energy of the light is related to its wavelength through $E_{\text{Photon}} = |E_m - E_n| = \hbar\omega = 2\pi f\hbar = hf = \frac{hc}{\lambda}$ when the electron changes from state *m* to state *n*.

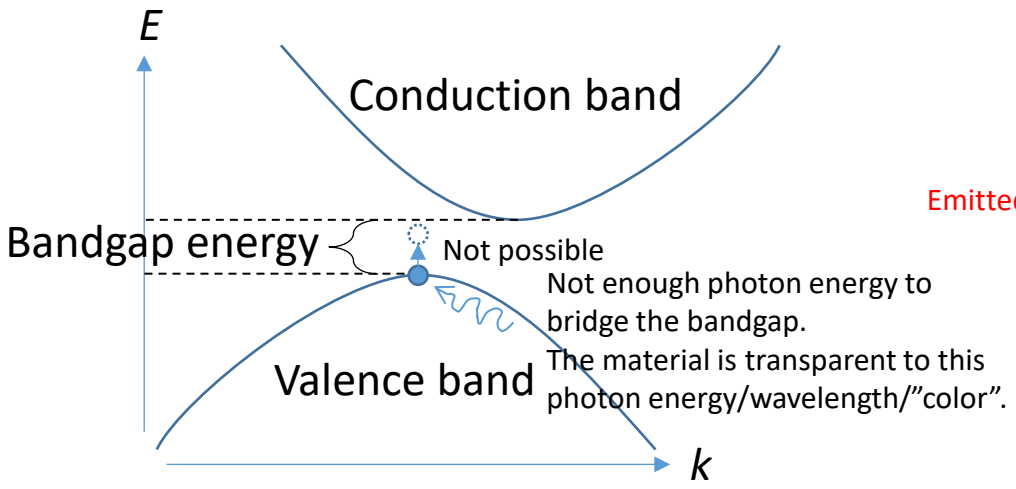


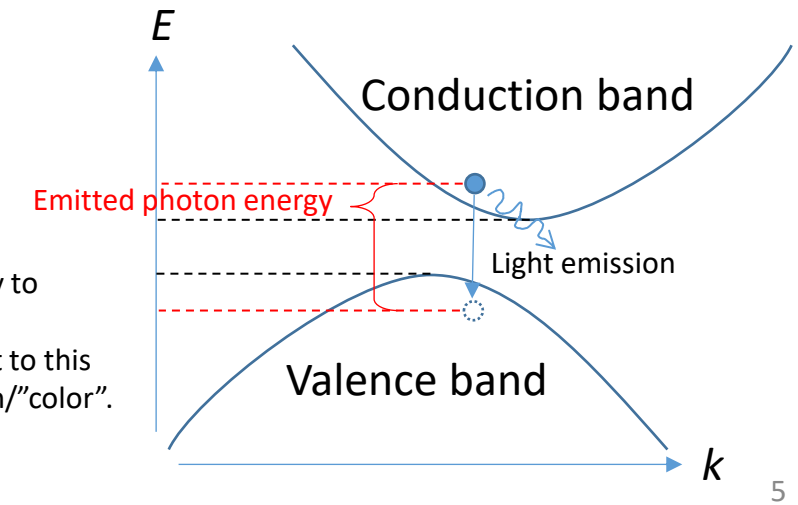


Light absorption and emission

- In materials, the outer (valence) electrons of the atoms are arranged in energy bands.
- In semi-conductors, the "valence band" is completely filled with electrons and the electrons needs a certain additional energy to jump into the "conduction band" (where they can move freely, e.g., to generate a current in a solar cell).
- The minimum energy needed to raise the electron from the valence band to the conduction band is named the bandgap.
- In Si the bandgap is 1.1 eV ~ $1.8 \cdot 10^{-19}$ J, meaning light with a wavelength <1100 nm has enough energy to move an electron from the valence band to the conduction band:
  - $E_{\text{Photon}} = \hbar\omega = \frac{hc}{\lambda}$, where $\hbar = \frac{h}{2\pi} = 1.05 \cdot 10^{-34}\ \text{J} \cdot \text{s} = 6.58 \cdot 10^{-16}\ \text{eV} \cdot \text{s}$ is the reduced Planck's constant.
  - $\hbar\omega = \frac{hc}{\lambda} = \frac{2\pi\hbar c}{\lambda} = \hbar c k \ \Rightarrow\ \omega = ck$ recovers the plane wave dispersion relation discussed earlier.
  - $E_{\text{Photon}} > 1.1\ \text{eV} \Rightarrow \hbar\omega = \frac{2\pi\hbar c}{\lambda} > 1.1\ \text{eV} \Rightarrow \lambda < \frac{2\pi\hbar c}{1.1\ \text{eV}} = 1126.8\ \text{nm}$
- Light with a wavelength >1100 nm is thus not absorbed in Si solar cells and does not generate any current.

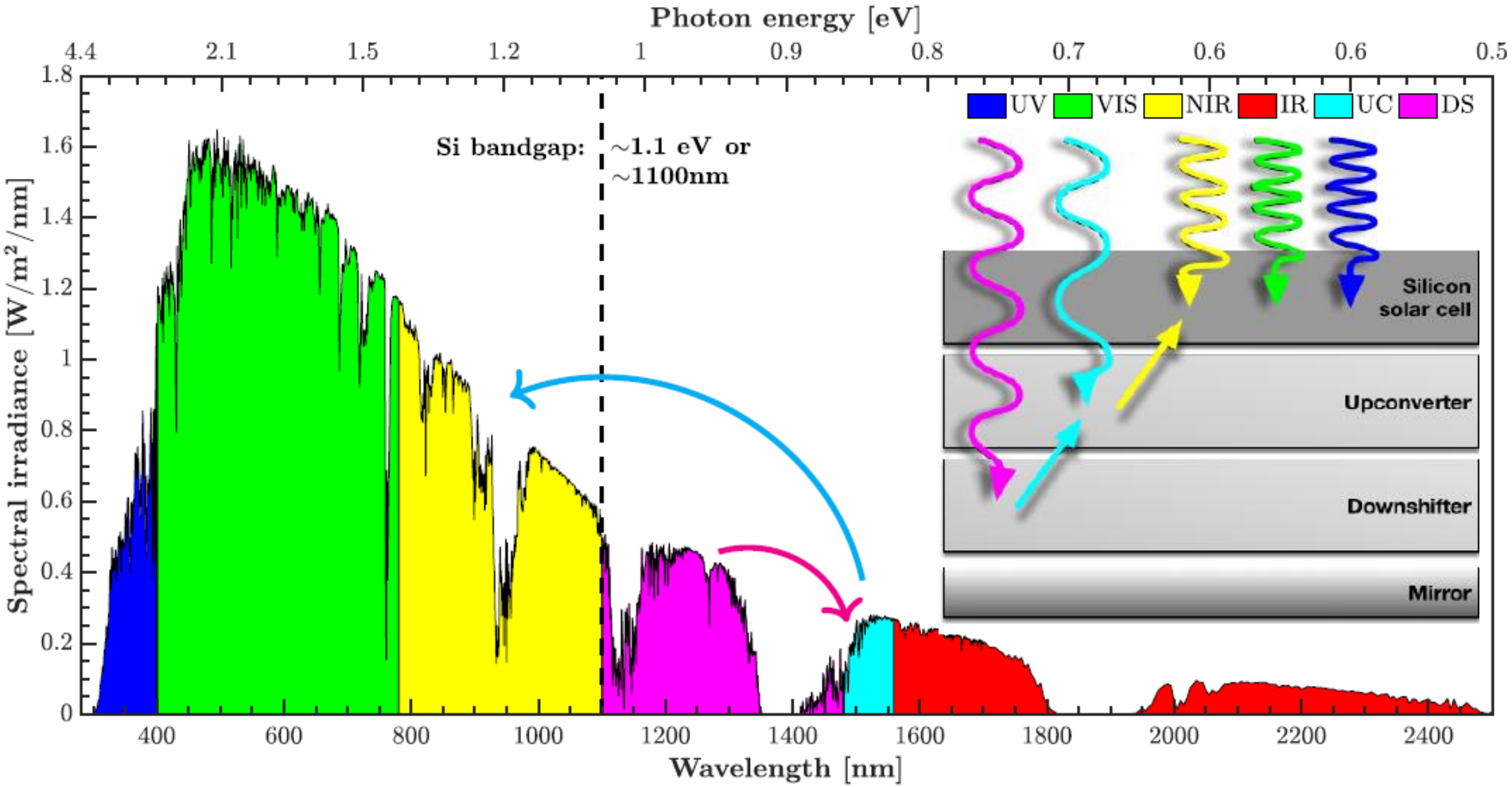






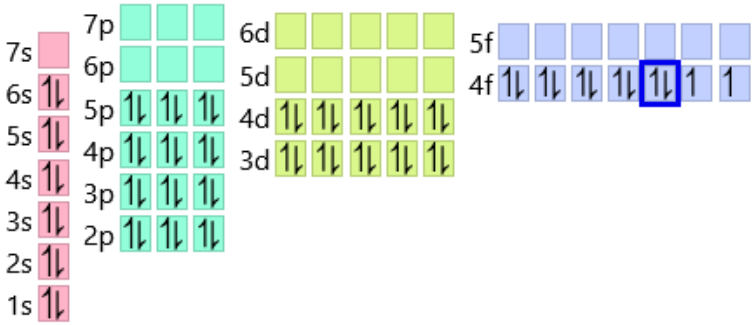
Upconversion

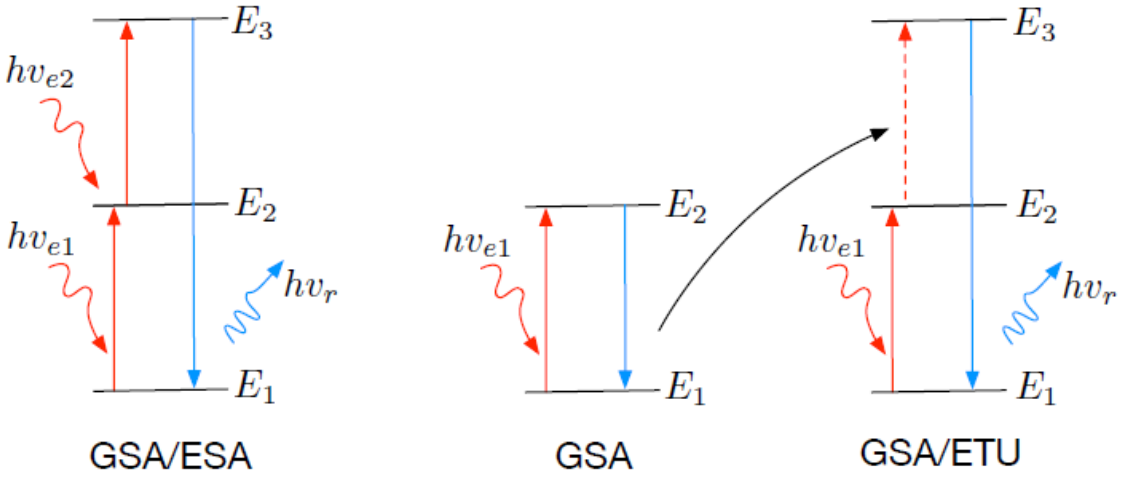
- The figure shows sunlight irradiance as a function of the wavelength.
- The >1100 nm parts of the sunlight are not absorbed in standard Si solar cells, because the photon energy is below the Si bandgap.
- Erbium ions can absorb a small part of the spectrum near 1500 nm and subsequently emit at <1100 nm (e.g., at 980 nm), which can be utilized in the Si cell.
- To utilize more of the spectrum, a downshifter material can absorb at 1100 nm - 1450 nm and emit at ~ 1500 nm.
- When utilizing downshifting, roughly 100 W/cm^2 is available to the upconverting Erbium ions, corresponding to 10% of the available solar irradiance.



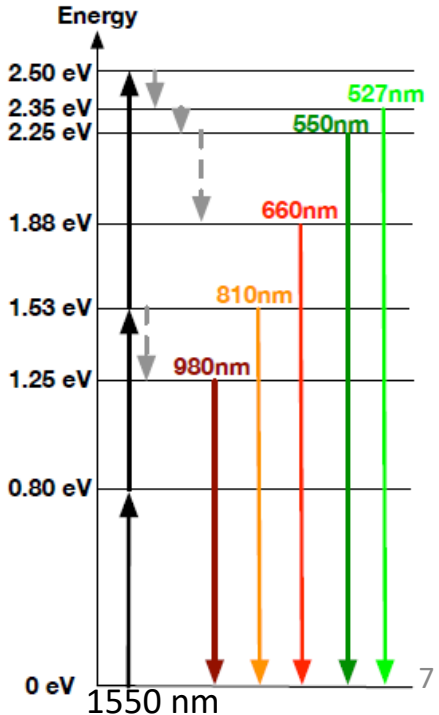
Upconversion in Erbium

- Exciting an atom twice, will ideally make the subsequent emitted light have twice the energy of the incoming photons (and half the wavelength).
- More excitation and decay routes are involved, and the situation is a bit more complicated => Described by rate-equation models!
- Rate equations describe the population of each energy level in a large collection of atoms, but we will not go into details with this.
- Erbium
  - 68 electrons.
  - Electrons in 4f shell is used for the upconversion.
  - Light around 1500 nm wavelength (infrared) is absorbed and subsequently emitted as <980 nm light (near-infrared).
  - In our case, the Erbium ions are doped into a TiO_2 matrix to create Er:TiO_2.



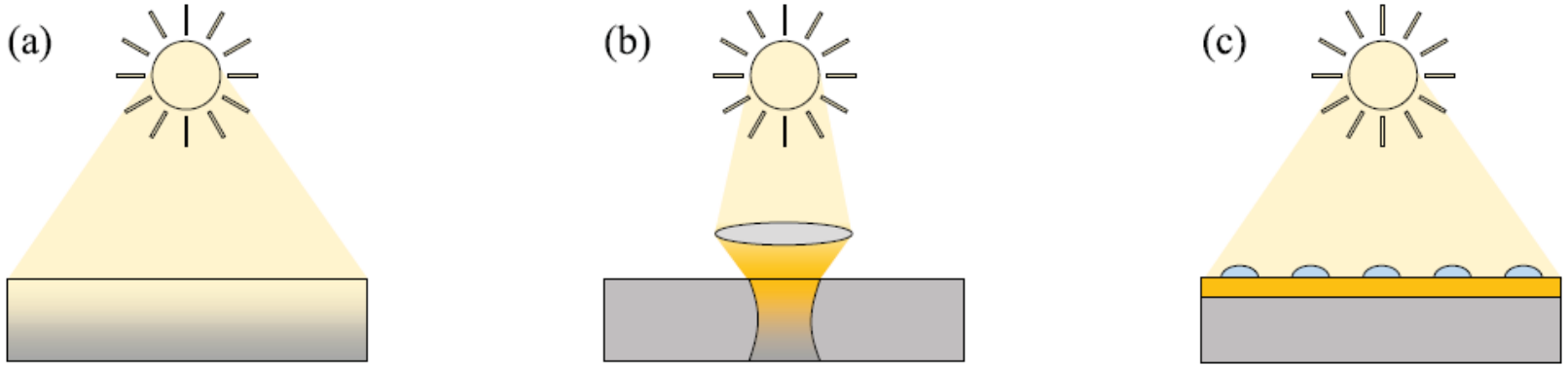


Er^{3+} 4f energy levels.



Enhancing the upconversion yield

- One can enhance the upconversion yield by concentrating the sunlight with wavelength ~1500 nm in an upconverting thin film.
  - This creates a higher density of photons, which can excite the electrons in the erbium atoms.
- One can concentrate the light using, e.g., an external lens but this may be inconvenient for many practical applications.
- Metal nanoparticles can provide both scattering and plasmonic enhancement of the electric field.
- The plasmonic enhancement can increase the field locally by a very large factor.
- The scattering redirects the light and can enhance the intensity by interference or by coupling to other modes of the combined system.
- Both the plasmonic enhancement and the scattering depend on the geometric shape of the nanoparticles.
- Scattering and absorption of light is described by Maxwell's equations.
- An optimized geometry can be found by topology optimization.



Recap: Maxwell's Equations

Maxwell's equations in matter are

Gauss Law, No Name, Faraday's law, Amperes law (with Maxwell correction)

$$\nabla \cdot \boldsymbol{D} = \rho_f \quad , \quad \nabla \cdot \boldsymbol{B} = 0 \quad , \quad \nabla \times \boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t} \quad , \quad \nabla \times \boldsymbol{H} = \boldsymbol{J}_f + \frac{\partial \boldsymbol{D}}{\partial t}.$$

$\boldsymbol{E}$ is the electric field, $\boldsymbol{D}$ is the electric displacement field, $\boldsymbol{B}$ is the magnetic field and $\boldsymbol{H}$ is the H-field (also named the magnetic field by some...). ρ_f are "free" charges and $\boldsymbol{J}_f$ are free currents. Often both are zero.

To close the equations, constitutive relations are needed. Linear isotropic non-magnetic materials satisfy

$$\boldsymbol{D} = \epsilon \boldsymbol{E} = \epsilon_0 \left(\epsilon' - j\epsilon''\right) \boldsymbol{E} \quad , \quad \mu_0 \boldsymbol{H} = \boldsymbol{B},$$

where $\epsilon = \epsilon_0 \epsilon_r$ and ϵ_r is the complex relative permittivity. It is related to the complex refractive index by

$$N^2 = (n - i\kappa)^2 = \epsilon_r = \frac{c_0^2}{c^2} \quad , \quad c_0 = 1/\sqrt{\mu_0 \epsilon_0} \approx 3 \cdot 10^8 \text{ m/s} \quad , \quad c = 1/\sqrt{\mu_0 \epsilon_r \epsilon_0}$$

The physics in ϵ (and N) is the response of the electrons in the material to an applied field.

If all electrons are tightly bound to the nuclei, the material is a dielectric (e.g. glasses) with $\kappa \approx 0$.

If some electrons can move easily, the material is a metal (e.g. gold) with $\kappa \neq 0$.

The $\kappa \neq 0$ leads to loss of energy in the wave when propagating in the material and high reflections (metals are "shiny").

Recap: Maxwell's Equations

Assuming

$$\boldsymbol{E}(\boldsymbol{r},t) = \boldsymbol{E}(\boldsymbol{r})e^{i\omega t} \quad , \quad \boldsymbol{B}(\boldsymbol{r},t) = \boldsymbol{B}(\boldsymbol{r})e^{i\omega t} \quad , \quad \boldsymbol{J}_f(\boldsymbol{r},t) = \boldsymbol{J}_f(\boldsymbol{r})e^{i\omega t},$$

and inserting into the electric vector wave equation

$$\mu_0 \frac{\partial^2 \epsilon \boldsymbol{E}}{\partial t^2} + \nabla \times (\nabla \times \boldsymbol{E}) = -\mu_0 \frac{\partial \boldsymbol{J}_f}{\partial t}$$

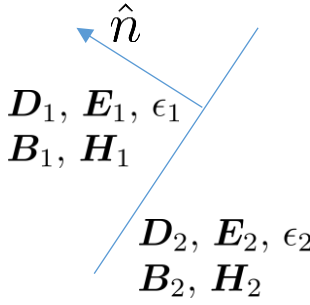
yield the time-harmonic vector wave equation for the electric field:

$$-\nabla \times (\nabla \times \boldsymbol{E}) + \frac{\omega^2}{c_0^2}\epsilon_r \boldsymbol{E} = i\omega\mu_0 \boldsymbol{J}_f.$$

The electric field can oscillate in different directions, which is known as the polarization.

Recap: Maxwell's Equations

From Maxwell's equations, one can derive so-called interface conditions, for looking at what happens to the fields across a material interface:



$$\hat{n}\cdot(\boldsymbol{D}_1-\boldsymbol{D}_2)=\sigma_f \quad , \quad \hat{n}\cdot(\boldsymbol{B}_1-\boldsymbol{B}_2)=0 \quad , \quad \hat{n}\times(\boldsymbol{E}_1-\boldsymbol{E}_2)=0 \quad , \quad \hat{n}\times(\boldsymbol{H}_1-\boldsymbol{H}_2)=\boldsymbol{K}_f$$

For $\sigma_f=0$ and $\boldsymbol{K}_f=0$, the 1st and 3rd equations says: $\hat{n}\cdot(\epsilon_1\boldsymbol{E}_1)=\hat{n}\cdot(\epsilon_2\boldsymbol{E}_2)$ and $\hat{n}\times\boldsymbol{E}_1=\hat{n}\times\boldsymbol{E}_2$. Writing

$$\boldsymbol{E}_1=\boldsymbol{E}_1^{||}\hat{n}+\boldsymbol{E}_1^{\perp}\hat{m}_1 \quad , \quad \boldsymbol{E}_2=\boldsymbol{E}_2^{||}\hat{n}+\boldsymbol{E}_2^{\perp}\hat{m}_2$$

with $\hat{m}_i$ perpendicular to $\hat{n}$, it is seen that normal component of the electric field is allowed to jump and the tangential component of the electric field is continuous across a material interface.

These interface conditions leads to the fact, that Lagrange finite element does not work, in general, for electromagnetic wave problems, since they create piecewise continuous functions.

Instead one uses so-called Nédélec elements, which are created for vector fields where only the tangential component (along mesh element boundaries) are continuous.

Nédélec elements are also available in fenics.

They are also named $H(\text{curl})$-conforming finite elements.

Finite Element Modeling

Nedelec or edge elements are finite element shape functions for constructing functions in H(curl).

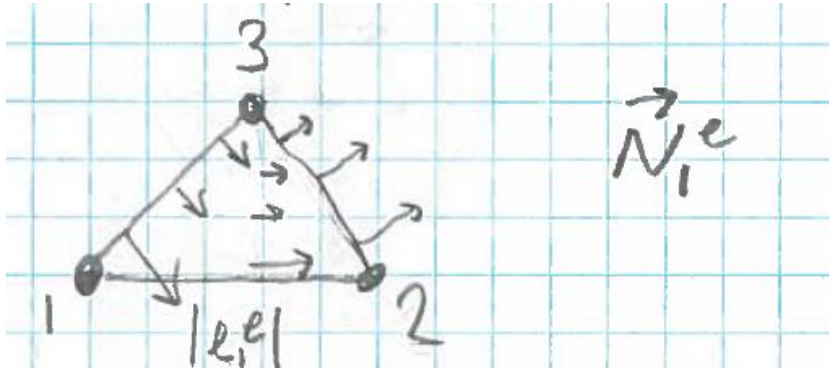
Instead of assigning the degrees of freedom to nodal values, they are assigned to edges.

Mesh elements sharing an edge will agree on the quantity on the edge, and the quantity is thus continuous across edges and faces.

In 2D, let $\vec{N}_1^e = \frac{r}{h_3^e}\hat{\theta}$, $\hat{\theta} = (\sin\theta, \cos\theta)$, where r and θ are polar coordinates centered on node 3 and h_3^e is the shortest distance from node 3 to the edge 1-2.

$\vec{N}_1^e$ will not have a tangential component on the two edges originating in node 3.

On edge 1-2 there will be a tangential component, which is the degree of freedom.



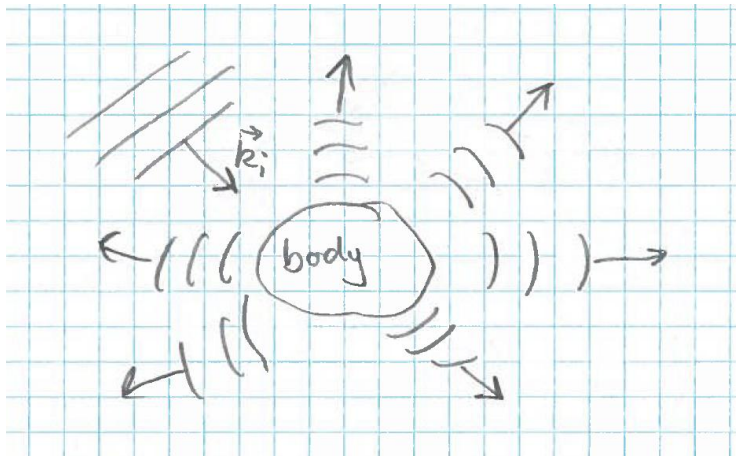
Scattered Field

Implicit definition of the source term is done by splitting the total field $\boldsymbol{E}$ as

$$\boldsymbol{E} = \boldsymbol{E}_i + \boldsymbol{E}_s$$

where $\boldsymbol{E}_i$ is the incident field: $\vec{E}_i$ solves the wave equation without the scatterer present.

$\boldsymbol{E}_s$ is the scattered field.



This yield for $\boldsymbol{J}_f = 0$

$$-\nabla \times (\nabla \times \boldsymbol{E}) + \frac{\omega^2}{c_0^2}\epsilon_r \boldsymbol{E} = 0 \Rightarrow -\nabla \times (\nabla \times \boldsymbol{E}_s) + \frac{\omega^2}{c_0^2}\epsilon_r \boldsymbol{E}_s = \nabla \times (\nabla \times \boldsymbol{E}_i) - \frac{\omega^2}{c_0^2}\epsilon_r \boldsymbol{E}_i$$

to be solved for $\boldsymbol{E}_s$. The incident field implicitly defines the current density generated by the scatterer.

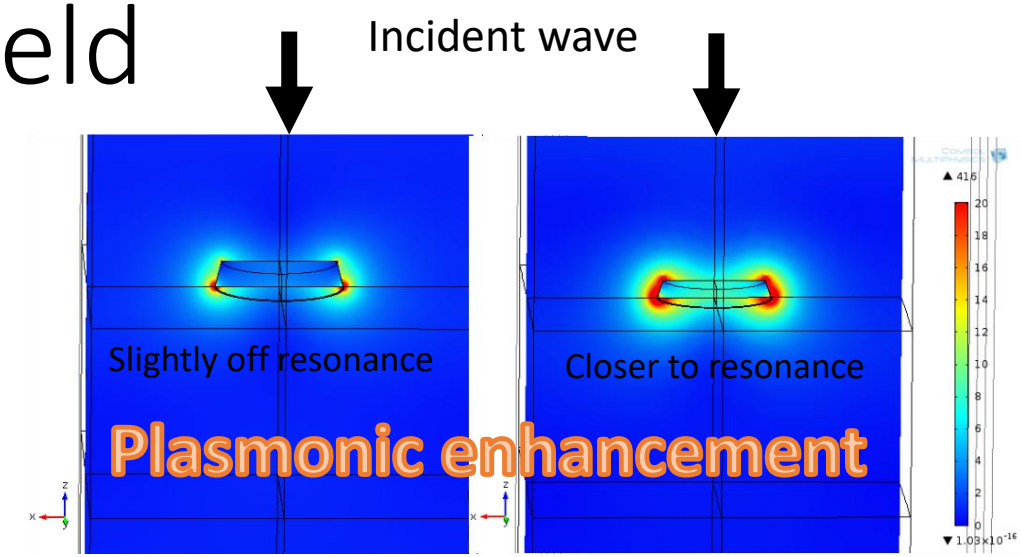
This allows one to calculate the scattering properties of the system.

Remember that in 2D transverse electric case, the double-curl equation reduces to the equation from acoustics (Helmholtz equation).

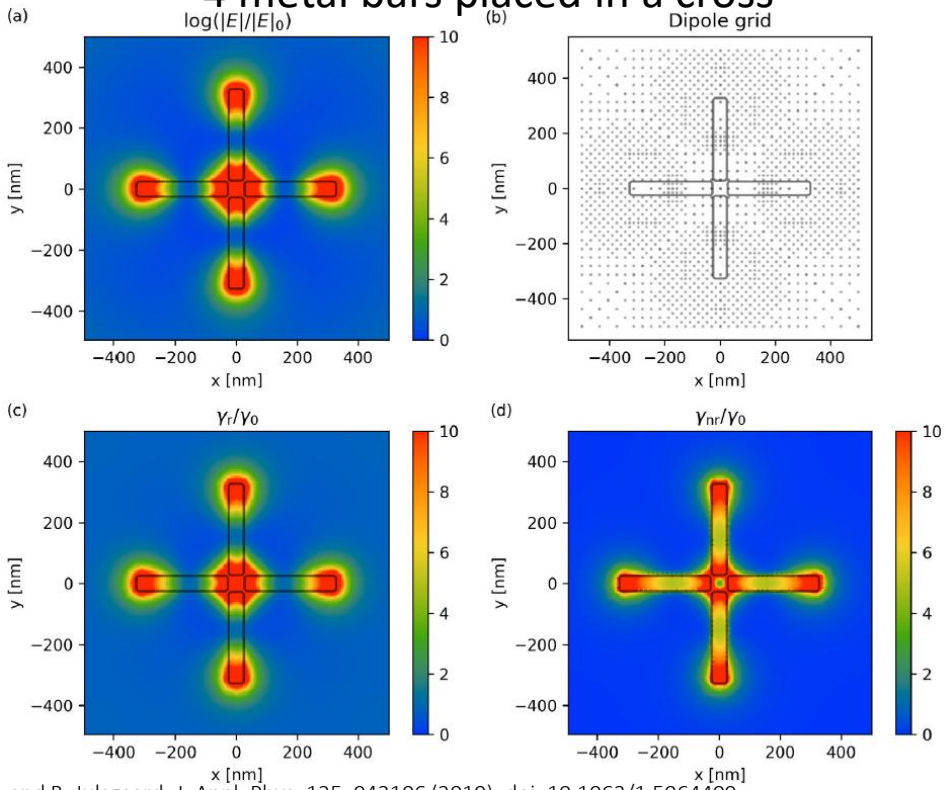
This scattered field method thus also works for acoustic problems. It is also knows as the background field method.

Enhancing the upconversion yield

- Plasmonic enhancement can provide very high fields in localized volumes.
- Plasmonic enhancement happens when the external field is in resonance with the conduction electrons in a metal.
- Expected to be relatively independent of the angle of the incoming light.
- This is perfect for a the upconversion process, which scales as the intensity squared, I^2, for sunlight intensity, $I \ll I_{sat}$, where I_{sat} is a material property related to upconversion.
- The high fields are usually in close proximity to the metal nanoparticle.
- Erbium ions placed close to a metal particle will have a large non-radiative decay rate – some of the upconverted light will be absorbed in the metal and not "radiated far away".
- This may limit the applicability of plasmonics to enhance upconversion.

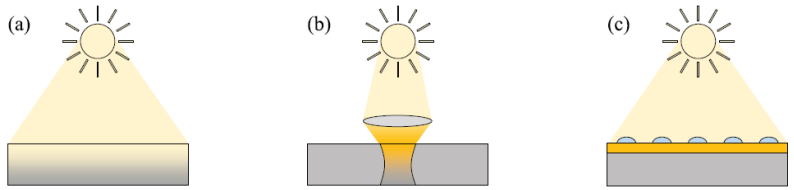


4 metal bars placed in a cross

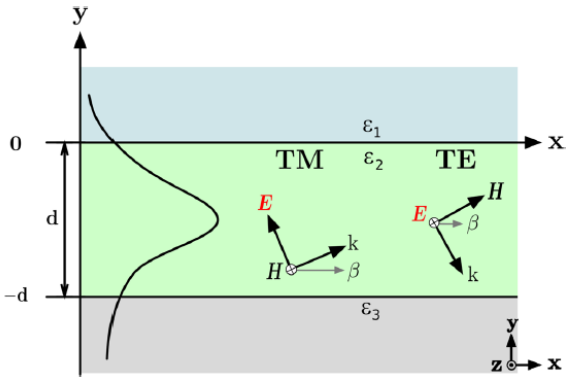


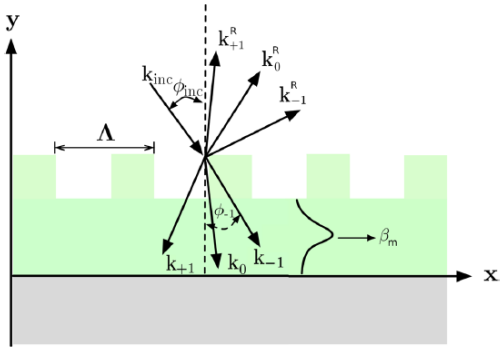
J. Christiansen, H. Lakhotiya, E. Eriksen, S. P. Madsen, P. Balling, and B. Julsgaard, J. Appl. Phys. 125, 043106 (2019); doi: 10.1063/1.5064409

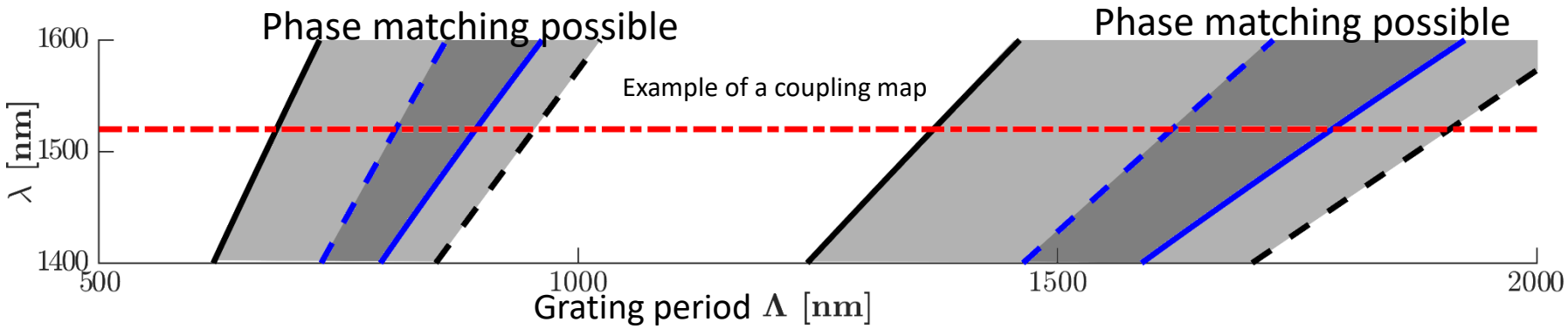
Enhancing the upconversion yield



- Thin films with higher refractive index than the surrounding materials can support wave guide modes, where a wave is travelling in the film plane.
- A grating is a periodic material-pattern and will diffract an incoming wave in different directions.
- If one of the diffraction grating models match up with a waveguide mode, the system can efficiently couple incoming light into a wave traveling in the film plane.
- This will increase the intensity of light in the whole volume of the film.
- Expected to depend a lot on the angle of the incoming light.
- (Semi-)Analytic wave guide analysis gives the conditions for phase matching the incoming light to excite guided mode in film.







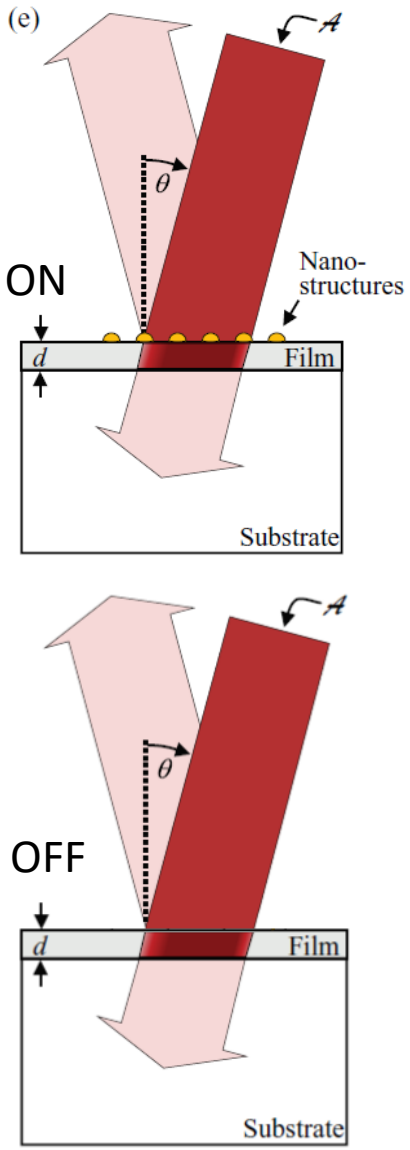
J. Vester-Petersen, S. P Madsen, O. Sigmund, P. Balling, B. Julsgaard, R. E Christiansen, Field-enhancing photonic devices utilizing waveguide coupling and plasmonics-a selection rule for optimization-based design, Optics Express 26, 18, A788-A795 (2018)

Enhancing the upconversion yield

- Consider now two samples: One containing nanostructures and another one without nanostructures.
- To quantify the enhancement of the upconverted light due to the nanostructures, we define

$$L_{\mathrm{UCL}} = \frac{Y_{\mathrm{UCL}}^{\mathrm{on}}}{Y_{\mathrm{UCL}}^{\mathrm{off}}} = \frac{\int_{\mathcal{V}_{\mathrm{UC}}} f^{(\mathrm{b})}\left(\frac{|\boldsymbol{E}|^2}{|E_{\mathrm{sat}}|^2}\right) \mathrm{d}V}{\int_{\mathcal{V}_{\mathrm{UC}}} f^{(\mathrm{b})}\left(\frac{|\boldsymbol{E}_b|^2}{|E_{\mathrm{sat}}|^2}\right) \mathrm{d}V}, \qquad (101) \quad f^{(\mathrm{b})}(\bar{I}/I_{\mathrm{sat}}) = 1 + \ln\left(\frac{\sqrt{1+2\bar{I}/I_{\mathrm{sat}}}+1}{2}\right) - \sqrt{1+2\bar{I}/I_{\mathrm{sat}}} + \frac{\bar{I}}{2I_{\mathrm{sat}}}, \qquad (23)$$

  - "on" means the beam is on the part of the sample with nanostructures.
  - "off" means the beam is off the part of the sample with nanostructures, i.e., on the part without nanostructures.
  - $Y_{\mathrm{UCL}}^{\mathrm{on}}$ and $Y_{\mathrm{UCL}}^{\mathrm{off}}$ can be measured directly.
- The integrals can be calculated numerically:
  - E is the electric field when nanostructures are present.
  - E_b is the electric field without the nanostructures.
  - E_{sat} is the electric field for the conditions where I_{sat} was measured.
- E_b and E_{sat} can in our case be calculated analytically using the transfer matrix method.
- E, in the presence of nanostructures, must be calculated numerically.



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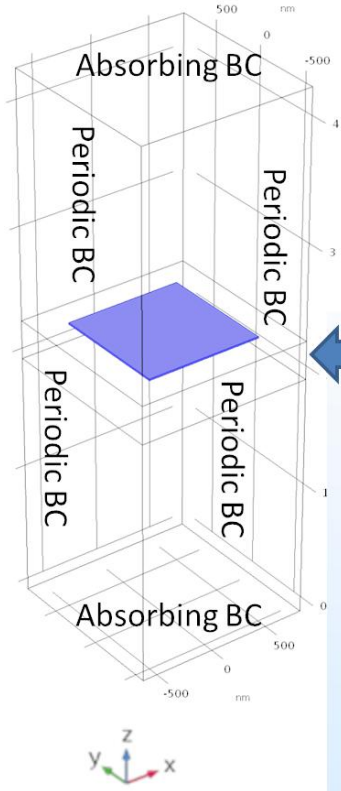
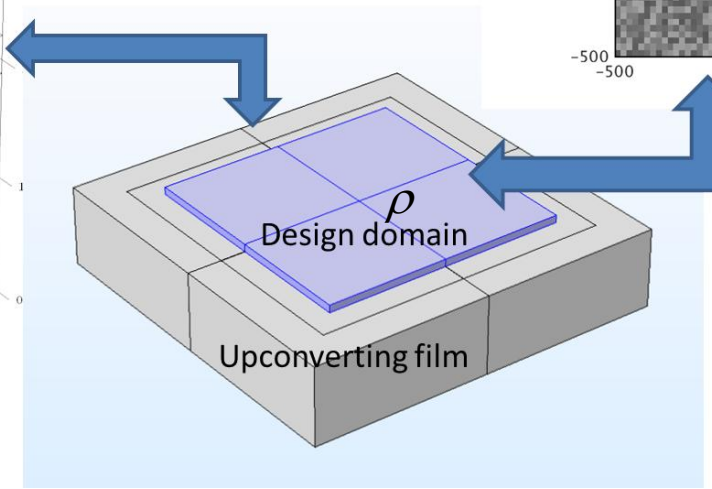
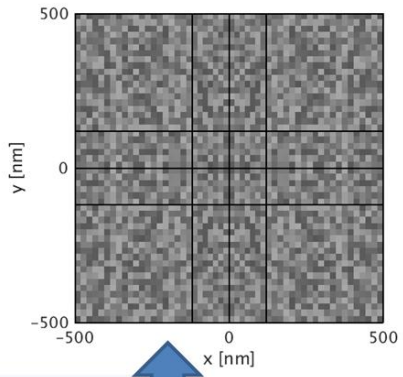
Topology Optimization

- Since the enhancing effects depends on the nanoparticle geometry, we formulate an optimization problem to determine an optimal geometry.
- This is formulated in a (numerical) design density function, ρ.
- ρ is a taken to be a 2D piecewise constant function of space and extruded into 3D.
- The material is specified through the relative permittivity, ε_r., in the governing equation $-\nabla \times (\nabla \times \boldsymbol{E}) + \frac{\omega^2}{c_0^2}\varepsilon_r \boldsymbol{E} = i\omega\mu_0 \boldsymbol{J}_f$.
- In the "design domain", ε_r becomes a function of the density function, $\varepsilon_r(\boldsymbol{r})=\varepsilon_r(\rho(\boldsymbol{r}))$, s.t.
  - $\varepsilon_r(\rho=1)=\varepsilon_{r,Au}$
  - $\varepsilon_r(\rho=0)=\varepsilon_{r,Air}$.
- ρ implicitly determines the spatial distribution of Au and Air.
- A non-linear interpolation scheme is used for densities between 0 and 1:
  - $\varepsilon_r = (n^2 - \kappa^2) - 2in\kappa$
  - $n = n_{\text{Air}} + \rho(n_{\text{Au}} - n_{\text{Air}}),\ \kappa = \kappa_{\text{Air}} + \rho(\kappa_{\text{Au}} - \kappa_{\text{Air}})$
- The interpolation scheme must not introduce artificial plasmonic resonances!
- The optimization problem will maximize the upconversion yield by changing the geometry of the Au structure on top of the upconverting Er:TiO_2 film.





Top view of ρ initial "design".



A non-linear material interpolation for design of metallic nano-particles using topology optimization, R. E. Christiansen, J. Vester-Petersen, S. P. Madsen, O. Sigmund Computer Methods in Applied Mechanics and Engineering 343, 23-39 (2019)

Topology Optimization

- The objective function may include: Multiple design realizations (geometric robustness), multiple frequencies, multiple angles of incidence, and multiple polarizations:

$$\Phi(\rho) = \frac{1}{2} \sum_{TE,TM} \frac{\int_{\Omega_T} |E(\tilde{\boldsymbol{\rho}}_i, \omega_j, \phi_k)|^3 d\Omega}{\int_{\Omega_T} |E(\mathbf{0}, \omega_j, \phi_k)|^3 d\Omega}$$

- $|E|^3$ dependence is chosen to optimize for "intermediate" incoming field strength in

$$L_{\mathrm{UCL}} = \frac{Y_{\mathrm{UCL}}^{\mathrm{on}}}{Y_{\mathrm{UCL}}^{\mathrm{off}}} = \frac{\int_{\mathcal{V}_{\mathrm{UC}}} f^{(\mathrm{b})}\left(\frac{|\boldsymbol{E}|^2}{|E_{\mathrm{sat}}|^2}\right) \mathrm{d}V}{\int_{\mathcal{V}_{\mathrm{UC}}} f^{(\mathrm{b})}\left(\frac{|\boldsymbol{E}_b|^2}{|E_{\mathrm{sat}}|^2}\right) \mathrm{d}V}, \qquad (101)$$

- The optimization problem is

$$f^{(\mathrm{b})}(\bar{I}/I_{\mathrm{sat}}) = 1 + \ln\left(\frac{\sqrt{1+2\bar{I}/I_{\mathrm{sat}}}+1}{2}\right) - \sqrt{1+2\bar{I}/I_{\mathrm{sat}}} + \frac{\bar{I}}{2I_{\mathrm{sat}}}, \qquad (23)$$

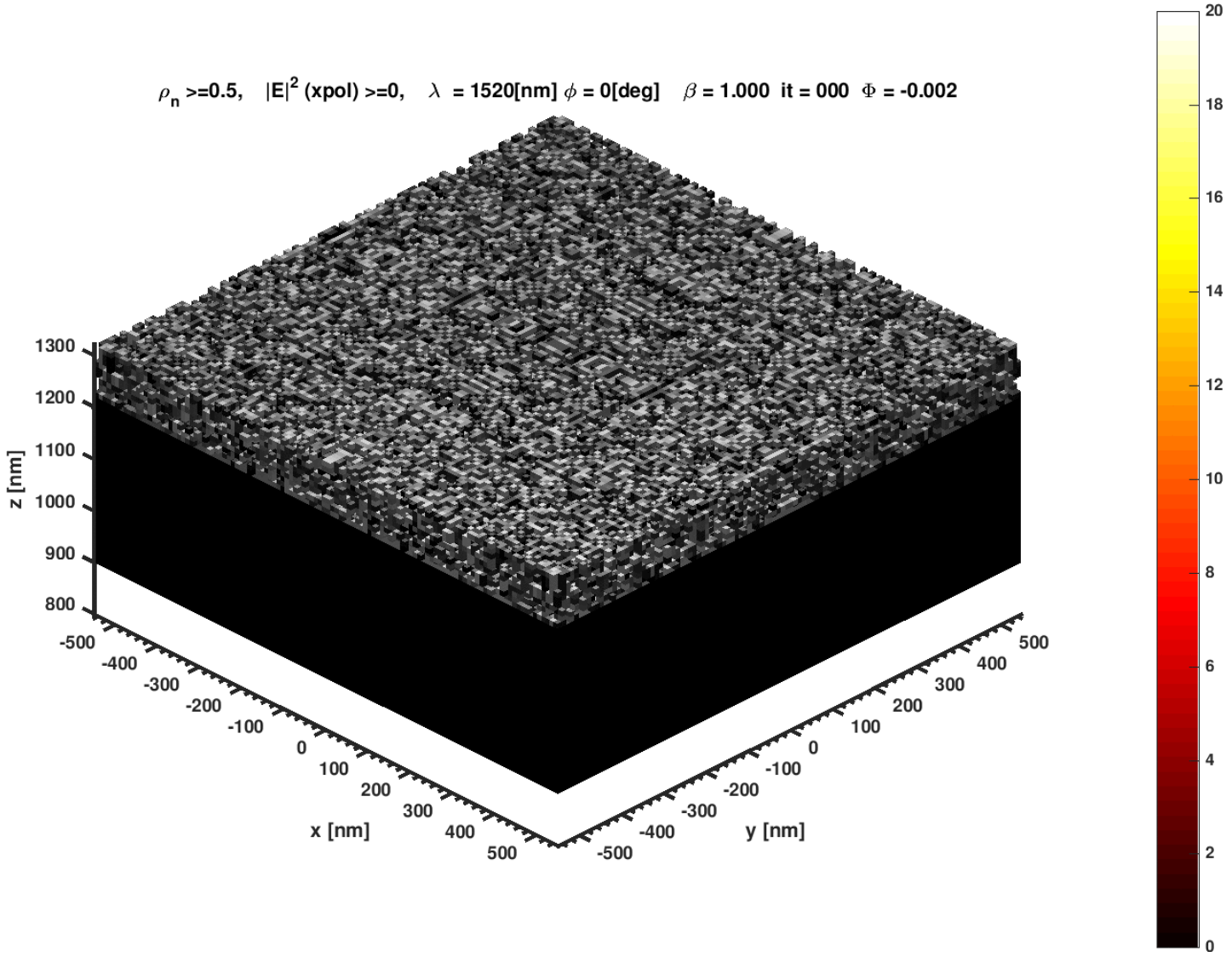
$$\begin{aligned} \text{maximize} \ \ & \min_{i,j,k} \Phi(\rho) \text{ with respect to } \rho \\ \text{s.t.} \ \ & 0 \leq \rho(\boldsymbol{r}) \leq 1 \ , \\ & \epsilon_{r,i} = \epsilon_r(\tilde{\boldsymbol{\rho}}_i) \ , \\ & \tilde{\boldsymbol{\rho}}_i = \mathrm{Project}_i(\eta_i, \mathrm{Filter}_i(\boldsymbol{\rho})) \end{aligned}$$

where i numbers the geometric realizations, j the frequencies, and k the incoming angles.

- The optimization problem maximizes the worst enhancement of all realizations parametrized by i, j and k. This ensures that none of the cases perform poorly.

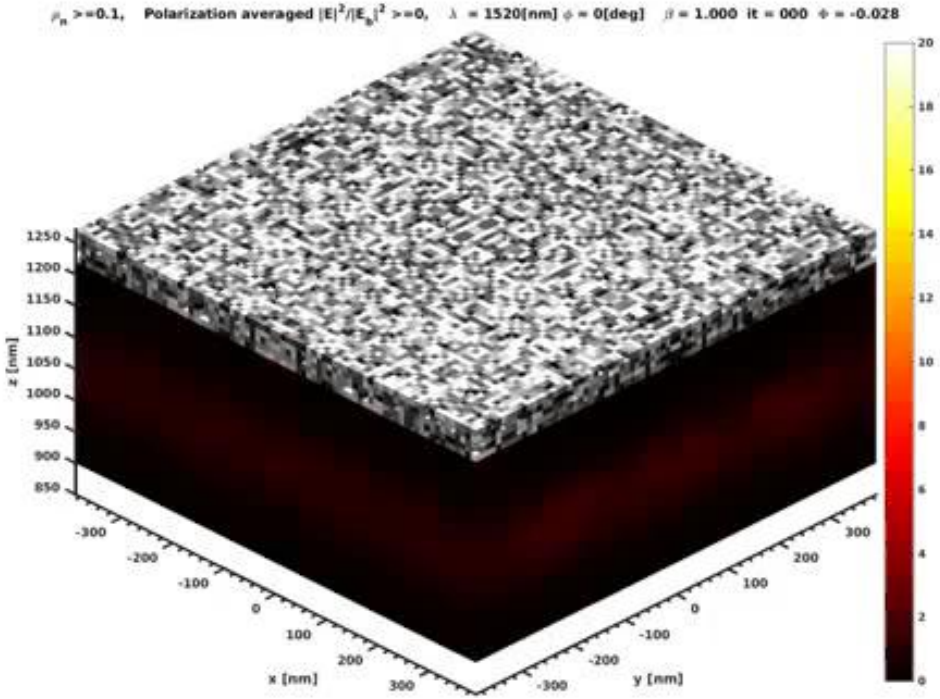
- We often chose to average over the polarization, but one could also include the polarizations in the max-min optimization.

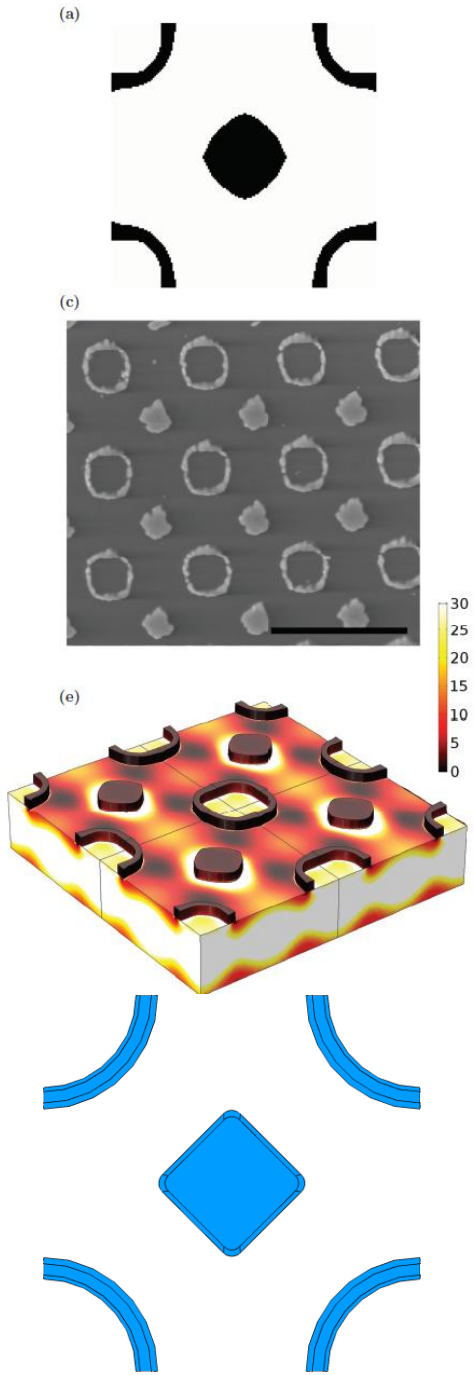
Topology Optimization Iterations



Experiments

Another design that excites a waveguide mode:





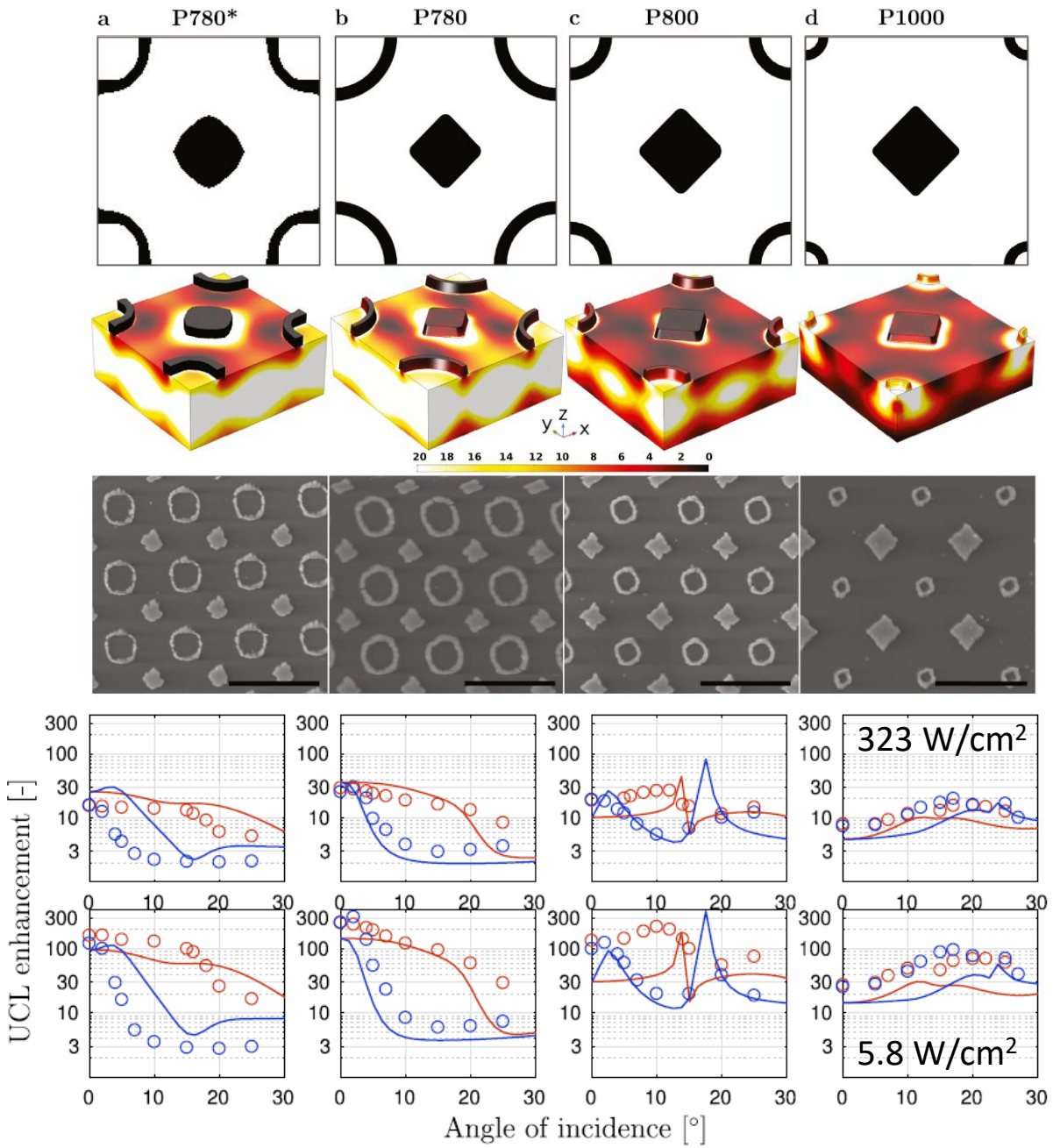
Simplification and parametric re-optimization:

- To make EBL production easier!
- To be able to investigate more easily the influence of the different parts of the geometry, e.g., the right thickness.
- Direct production via EBL from the topology optimized design was also done.

J. Christiansen, J. Vester-Petersen, S. Roesgaard, S.H. Møller, R. E. Christiansen, O. Sigmund, S.P. Madsen, P. Balling and B. Julsgaard, Solar Energy Materials and Solar Cells 208, 110406 (2020)

Experiments

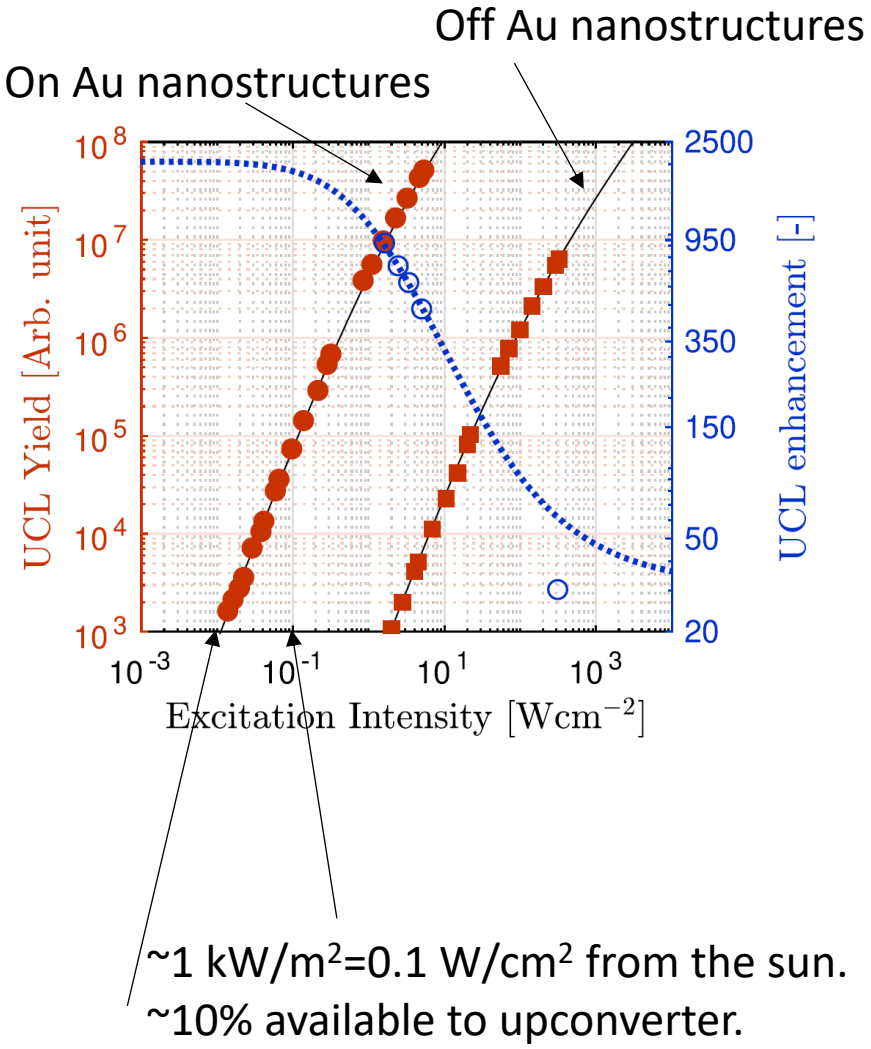
- Several realizations of the same general design were made.
- UC enhancement calculation fits reasonably well with measurements, both at high and low incoming intensity.
- Maximum enhancement of the upconversion at 5.8 W/cm^2 is roughly 300x.
- 913x enhancement at 1.7 W/cm^2 has since been measured.



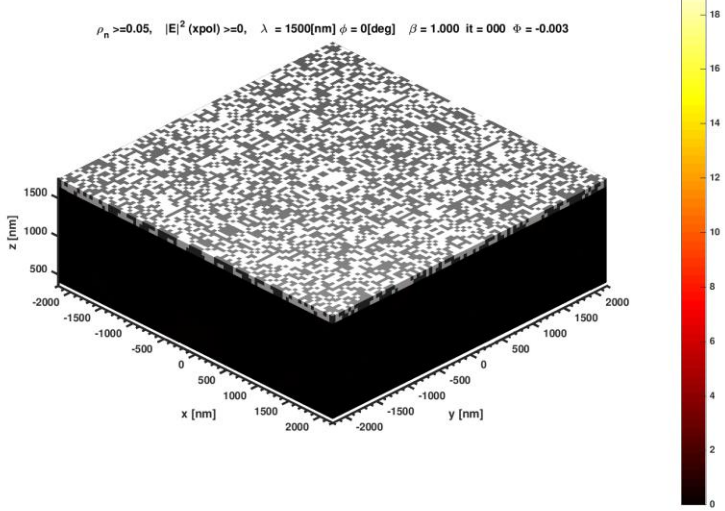
J. Christiansen, J. Vester-Petersen, S. Roesgaard, S.H. Møller, R. E. Christiansen, O. Sigmund, S.P. Madsen, P. Balling and B. Julsgaard, Solar Energy Materials and Solar Cells 208, 110406 (2020)

Experiments

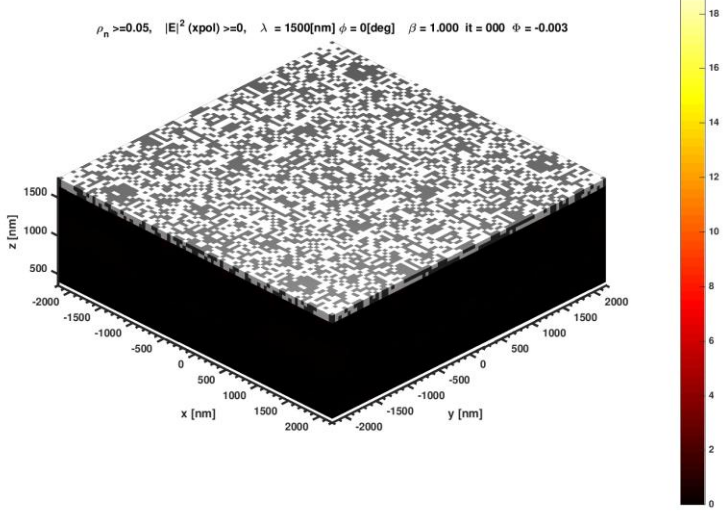
- Changing the incoming intensity changes the upconversion signal and the enhancement factor.
  - If $I << I_{sat}$, the upconversion yield scale as $(I/I_{sat})^2$.
  - If $I >> I_{sat}$, the upconversion yield scale as I/I_{sat}.
- Lowering the incoming intensity increases the enhancement factor.
- At sunlight intensity, the Au nanostructures will enhance the signal about 2000 times!
- But the signal is unfortunately quite weak in absolute numbers ☹
- Highest UC enhancement measured in the lab was 913x at 1.7 W/cm^2
- The way forward to incorporate upconversion into commercial product is to find materials with better upconverting properties:
  - A larger absorption cross section, to absorb much more of the incoming light.
  - A high refractive index, to enable efficient light confinement.
- Materials with higher absorption cross section exist, e.g., Er-doped $NaYF_4/NaLuF_4$ core/shell nanocrystal films.
- There is thus some hope that this can be made even better with modern materials engineering!



J. Christiansen, J. Vester-Petersen, S. Roesgaard, S.H. Møller, R. E. Christiansen, O. Sigmund, S.P. Madsen, P. Balling and B. Julsgaard, Solar Energy Materials and Solar Cells 208, 110406 (2020)

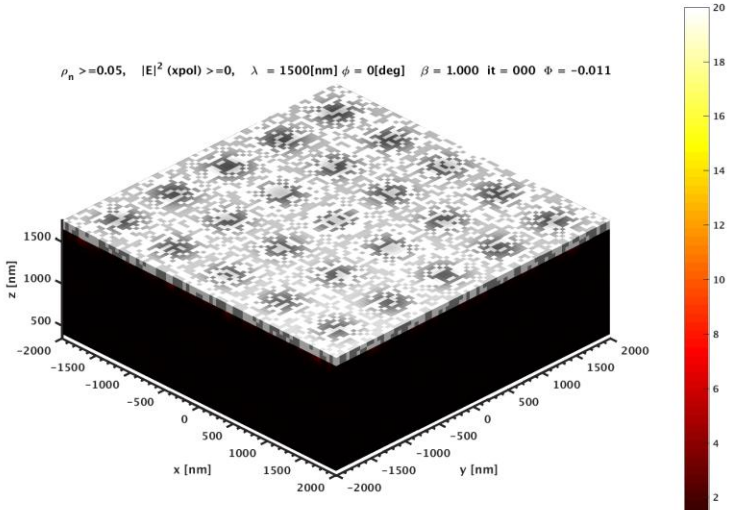


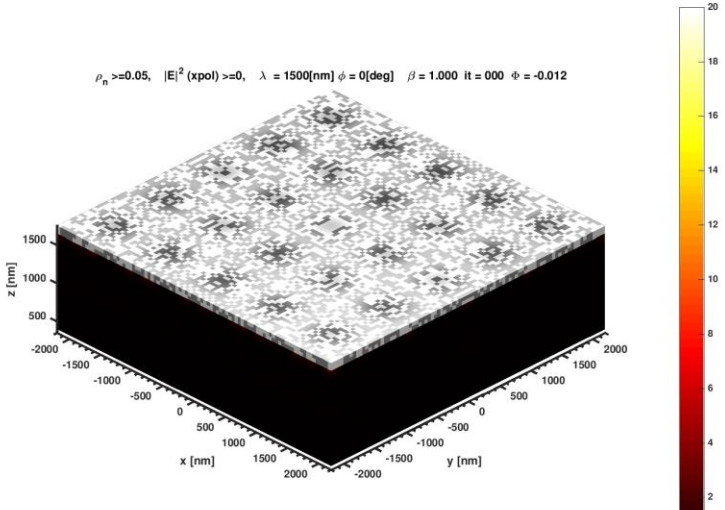
Conclusion



- We have developed a model, which describes upconversion in Erbium system reasonably well at both low and high intensities.
- We have developed a tool for obtaining optimized material distributions in electromagnetic wave propagation problems.
- Our application is upconversion but the methodology is useful for many other photonic problems.
- Topology optimization combined with Electron Beam Lithography allows for almost unlimited geometrical design freedom in 2D extruded nano structures.

- The upconversion model allows us to extrapolate the enhancement and upconversion yield at laboratory intensities to solar levels.
- Although the designed photonic structures was proved to be quite efficient, they alone were not enough to bring the upconversion yield to a scale suitable for commercial interests just yet.
- To further develop and commercialize upconversion for solar applications, one must first develop more efficient upconversion materials, e.g., with a high absorption cross section and high refractive index.



SunTune

suntune.au.dk